

SUYASH PRADHAN

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EDUCATION

Acharya Institute of Technology (VTU)

B.E. in Computer Science — GPA: 9.26 / 10.0

Kendriya Vidyalaya No. 3 (CBSE Class XII)

Science with PCM and Computer Science — 91%

Karnataka, India

Aug 2023 – Apr 2027

Punjab, India

May 2023

EXPERIENCE

Strategic Head

Bengaluru, Karnataka, India

Piston Club, Acharya Institute of Technology

Jan 2026 – May 2026

- Directed club-wide strategy across various verticals, coordinating a 50-member leadership team to deliver 10 technical events reaching 300+ student participants in 5 months.
- Designed quarterly growth roadmaps and accountability frameworks, aligning cross-team activities with the club's vision and increasing active member engagement by 40%.
- Led stakeholder communication with faculty advisors and external sponsors, securing resources and approvals for technical workshops and inter-college competitions.

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript, HTML, CSS

ML / AI: Scikit-learn, TensorFlow, PyTorch, Isolation Forest, MLP Autoencoder, One-Class SVM, Deep Learning, NLP, Generative AI, Prompt Engineering, FAISS, RAG

Cybersecurity: IDS/XDR, MITRE ATT&CK, Threat Intelligence, SOC Operations, Shannon Entropy Analysis, PGD Adversarial Hardening, SHAP/XAI, Network Security, Phishing Analysis, AI Security

Frameworks & Libraries: FastAPI, Flask, React.js, Tailwind CSS

Cloud & DevOps: AWS (EC2, S3, Lambda, CDK, Step Functions, DynamoDB, Textract, Comprehend, CloudWatch, X-Ray, CodePipeline), Docker, Git, GitHub, CI/CD, Agile/Scrum, Jira

Databases: MySQL, MongoDB, SQLite, DynamoDB

Tools: Jupyter Notebook, VS Code, Linux, Anaconda, Apache Kafka

Concepts: Software Architecture, SDLC, Data Structures & Algorithms, Computer Networking, IoT, UML, Data Engineering, Serverless Architecture

PROJECTS

AHRAS — Adaptive Hybrid Risk-Aware Security System

Python, Scikit-learn, FastAPI, Kafka, MongoDB | [GitHub](#)

Jan 2026 – Present

- Built a hybrid threat detection engine combining Isolation Forest, MLP Autoencoder, and One-Class SVM across 5 OCSF-normalised telemetry classes (network, host, cloud, process, file), achieving **F1 = 0.927** and **0% false positive rate** on labelled benchmarks using Python and Scikit-learn.
- Engineered a real-time 4-source telemetry pipeline (network packets, host agents, cloud audit logs, IoT) processing **376+ events/sec** via Apache Kafka, with Shannon entropy ransomware detection ($H > 7.2$ threshold), 23 MITRE ATT&CK-mapped signature rules, and AbuseIPDB threat intel enrichment.
- Deployed a FastAPI SOC platform with adversarially hardened ML (PGD augmentation, +12% evasion resistance), SHAP-style XAI explanations per alert, and formal adaptive risk scoring $R_t = w_1 S_{sig} + w_2 A_{ml}(1 + \Delta D) - w_3 T_{trust}$ at **26ms per-event latency**; outperforms signature-only baseline by **+6.2% F1**.

Enterprise Knowledge Intelligence System

Python, FAISS, Mistral-7B, Ollama, FastAPI | [GitHub](#)

Jan 2025 – Present

- Built a fully offline Version-Aware RAG system using FAISS HNSW and Mistral-7B (Ollama), supporting PDF/DOCX/TXT ingestion with **sub-50ms** retrieval latency across 10,000+ document chunks.
- Developed a 4-signal Weighted Hybrid Matching (WHM) ranking engine (semantic similarity, temporal decay, domain authority, lifecycle status), reducing Outdated Document Retrieval Rate (ODR) by **62%** over cosine-only baseline.

- Implemented 3-layer answer verification — Token-F1 faithfulness, embedding cosine grounding, and self-consistency sampling — with per-sentence citation traceability; evaluated across 7 IR metrics (P@K, R@K, NDCG@K, MRR, ODR, VA@1, latency).

Serverless Intelligent Document Processing Platform

AWS CDK, Lambda, Step Functions, DynamoDB, Textract | [GitHub](#)

Apr 2026 – Present

- Architected end-to-end serverless pipeline on AWS using API Gateway, Cognito JWT, S3, SQS + DLQ, and Lambda with full Infrastructure-as-Code via AWS CDK, KMS encryption at rest, and Cognito Advanced Security Mode.
- Orchestrated 4-state AI extraction via Step Functions, Textract (async OCR, multi-page PDF + table parsing), and Comprehend (entities, key phrases, sentiment) backed by a 2-table DynamoDB schema with 4 GSIs and Streams CDC processor for real-time index consistency.
- Delivered production observability with CloudWatch dashboards, 5 metric alarms, X-Ray tracing (5% baseline + 100% on errors), EMF structured log metrics, and full CodePipeline CI/CD (build → test → staging → manual approval → prod).

CERTIFICATIONS

Cybersecurity & AI Security

- Mastercard Cybersecurity Job Simulation — Forage *May 2026 ID: MJ8GgDq793hfcdfHS*
- Ethics in Cybersecurity; Cybersecurity Awareness: AI & Phishing Attacks — LinkedIn *Jan–Feb 2026*
- Introduction to AI Ethics and Safety — LinkedIn *Jan 2026*

Machine Learning & AI

- Deep Learning with Python: Optimizing Deep Learning Models — LinkedIn *May 2026*
- Python for Data Science and Machine Learning Essential Training — LinkedIn *May 2026*
- Machine Learning and AI Foundations — LinkedIn *Mar 2026*
- Principles of Gen AI; Prompt Engineering; AI Intelligence Primer Certification — Infosys Springboard *Jun 2025*
- Deep Learning; Computer Vision 101; NLP Basics — Infosys Springboard *Jun 2025*

Cloud & DevOps

- Next-Generation Cloud Computing & AWS Industry Readiness Program — Acharya Institute *Mar 2026*
- Introduction to Cloud Computing for IT Pros; Cloud Computing: Understanding Networking — LinkedIn *Jan–May 2026*

Software Engineering & Data

- Software Architecture: From Developer to Architect — LinkedIn *Jan 2026*
- Data Engineering Foundations — LinkedIn *May 2026*
- Software Development Life Cycle (SDLC); Agile Software Development — LinkedIn *Jan 2026*
- Programming Foundations: Algorithms & Databases — LinkedIn *Jan 2026*
- MySQL with Information Technology — Coursera *Nov 2024 ID: FXHWWTB7TYLS*

Other

- Renewable Energy Technology Fundamentals — University of Colorado Boulder *May 2026 ID: U9CE96PGV4FL*
- English for Career Development — University of Pennsylvania *Mar 2024*
- Work Smarter with Microsoft PowerPoint (with Honors) — Microsoft *Mar 2024*
- Solving Problems with Creative and Critical Thinking — IBM *Sep 2024 ID: KLAk9AESEN7G*